

Course Code : ECTH 81-1

KQLR/RS – 24 / 1283

**Eighth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

EVOLUTION OF AIR INTERFACE TOWARDS 5G

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Discuss role of wireless communication system in various domains. 5(CO1)
(b) Explore any five use cases of 5G. 5(CO1)
2. (a) Illustrate main principles of 5G design. 5(CO1)
(b) Explore any five parameters of 5G architecture system. 5(CO1)
3. (a) Discuss hardware technology for millimeter wave technology. 5(CO2)
(b) Illustrate any one beam scanning technique in millimeter wave technology. 5(CO2)
4. (a) Express significance of Massive Machine Type Communication (MTC). 5(CO3)
(b) What do you mean by Device to Device (D2D) Communication ? Illustrate your answer with an example. 5(CO3)

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Contd.

5. Discuss the features and utility of –

(i) Non Orthogonal Multiple Access (NOMA)

(ii) Sparse Code Multiple Access (SCMA). 10(CO4)

6. (a) Discuss how massive MIMO is implemented in LTE. 5(CO5)

(b) Illustrate MIMO configurations with the help of suitable diagram. 5(CO5)

